

narrow and congested that their utility for open spectrum is limited. Furthermore, the high frequency of the most prominent unlicensed bands limits signal propagation. Lower-frequency spectrum that penetrates weather, tree cover, and walls would provide significant advantages for services such as last-mile broadband connectivity.

The FCC should identify additional spectrum bands that can be designated for use as unlicensed “parks,” with a particular focus on frequencies below 2 GHz where propagation is best. The FCC will need to consult with other relevant agencies such as the Department of Defense, Federal Aviation Administration, and Department of Commerce; technical and scientific organizations such as the National Academy of Sciences and Institute of Electrical and Electronics Engineers; and the private sector. Furthermore, the U.S. government should work through the World Radio Conference and other international fora to create global unlicensed bands wherever possible.

There are many possible sources for additional unlicensed spectrum. The 5 GHz unlicensed band could be expanded relatively easily, a move that would also help bring the U.S. allocation in line with other countries. Because of its limited propagation characteristics, however, 5 GHz should not be seen as a long-term solution. Creating unlicensed spectrum parks elsewhere would involve relocation or other accommodation of existing users. That is a process the Commission has engaged in repeatedly in the past.

3. Remove Constraints On Underlay

The FCC took a major step forward with its February approval of ultra-wideband. The Commission wisely rejected overblown fears about interference, relying on technical data and prudent restrictions on UWB deployment. However, the Commission’s initial rules still put unnecessarily severe limits on where and how UWB can be used. Assuming that experience shows the fears about interference ungrounded, the FCC should loosen its restrictions without delay.